MATHEMATICS TESTING QUESTION BANK

Class XII - Question Compendium

SECTION A: MULTIPLE CHOICE QUESTIONS (MCQS)

Each question carries 1 mark. Select the most appropriate option from the choices given below.

Q1. If $|a| = 8$, $|b| = 3$ and $|a \times b| = 12$, then the value of $|a \cdot b|$ is:
(A) $6\sqrt{3}$
(B) $8\sqrt{3}$
(C) $12\sqrt{3}$
(D) $3\sqrt{12}$

Q2. If a square matrix A is such that $A^2 = A$ and $(I - A)^3 = xA + I$, then value of x must be:
(A) 7
(B) 5
(C) -7
(D) -1

Q3. The Cartesian equation of the line passing through the point (1, -3, 2) and parallel to the line $r = (2 + \lambda)i + \lambda j + (2\lambda - 1)k$ is:
(A) $(x - 1)/2 = (y + 3)/0 = (z - 2)/(-1)$
(B) $(x + 1)/1 = (y - 3)/1 = (z + 2)/2$
(C) $(x + 1)/2 = (y - 3)/0 = (z + 2)/(-1)$
(D) $(x - 1)/1 = (y + 3)/1 = (z - 2)/2$

Q4. The corner points of the feasible region in the graphical representation of an L.P.P. are (2, 72), (15, 20) and (40, 15). If $Z = 18x + 9y$ be the objective function, then:
(A) Z is maximum at (2, 72), minimum at (15, 20)
(B) Z is maximum at (15, 20), minimum at (40, 15)
(C) Z is maximum at (40, 15), minimum at (15, 20)
(D) Z is maximum at (40, 15), minimum at (2, 72)

Q5. The number of points of discontinuity of $f(x) = x + 3$, if $x \leq -3$; $-2x$, if $-3 < x < 3$; $6x + 2$, if $x \geq 3$ is:
(A) 0
(B) 1
(C) 2
(D) infinite

Q6. If E and F are two independent events such that $P(E) = 3/10$, $P(E \cup F) = 1/2$, then $P(E|F) - P(F|E)$ is equal to:
(A) 2/7
(B) 3/35
(C) 1/70
(D) 1/7

Q7. Given a curve $y = 7x - x^3$ and x increases at the rate of 2 units per second. The rate at which the slope of the curve is changing, when $x = 5$ is:
(A) -60 units/sec
(B) 60 units/sec
(C) -70 units/sec
(D) -140 units/sec

Q8. The common region determined by all the constraints of a linear programming problem is called:
(A) an unbounded region
(B) an optimal region
(C) a bounded region
(D) a feasible region

Q9. If A = [-1 0 0 ; 0 1 0 ; 0 0 1], then A^{-1} is:
(A) [-1 0 0 ; 0 -1 0 ; 0 0 -1]
(B) [1 0 0 ; 0 -1 0 ; 0 0 -1]
(C) [-1 0 0 ; 0 -1 0 ; 0 0 1]
(D) [-1 0 0 ; 0 1 0 ; 0 0 1]

Q10. If d/dx (f(x)) = log x, then f(x) equals:
(A) -1/x + C
(B) x(log x - 1) + C
(C) x(log x + x) + C
(D) 1/x + C

SECTION A: ASSERTION-REASONING QUESTIONS

For questions 11 to 16, two statements are given - one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the following options: (A) Both A and R are true and R is the correct explanation of A. (B) Both A and R are true, but R is not the correct explanation of A. (C) A is true, but R is false. (D) A is false, but R is true.

Q11. Assertion (A): Lines given by x = py + q, z = ry + s and x = p'y + q', z = r'y + s' are perpendicular to each other when pp' + rr' = 1.
Reason (R): Two lines $r = a_1 + \lambda b_1$ and $r = a_2 + \mu b_2$ are perpendicular to each other if $b_1 \cdot b_2 = 0$.

Q12. Assertion (A): One of the particular solutions of the differential equation dy/dx = e^(x + y) can be e^x + e^(-y) = -2.
Reason (R): e^x + e^(-y) = C is the general solution of the differential equation dy/dx = e^(x + y).

Q13. Assertion (A): The vectors a = 6i + 2j - 8k, b = 10i - 2j - 6k and c = 4i - 4j + 2k represent the sides of a right angled triangle.
Reason (R): Three non-zero vectors of which none of two are collinear form a triangle if their resultant is zero vector or sum of any two vectors is equal to the third.

Q14. Assertion (A): For any symmetric matrix A, B'AB is a skew-symmetric matrix.
Reason (R): A square matrix P is skew-symmetric if P' = -P.

Q15. Assertion (A): The range of the function $f(x) = 2 \sin^{-1} x + 3\pi/2$, where $x \in [-1, 1]$, is $[\pi/2, 5\pi/2]$.
Reason (R): The range of the principal value branch of $\sin^{-1}(x)$ is $[0, \pi]$.

Q16. Assertion (A): f(x) = 3x - 8, x ≤ 5; 2k, x > 5 is continuous at x = 5 for k = 5/2.
Reason (R): For a function f to be continuous at x = a, lim x→a- f(x) = lim x→a+ f(x) = f(a).

SECTION B: VERY SHORT ANSWER QUESTIONS (VSAQS)

Each question carries 2 marks. Answer the following questions precisely with necessary formulas or clear steps.

Q17. If the lines (x - 3)/1 = (1 - y)/1 = (z + 2)/p and (2 - x)/3 = (y + 1)/5 = (z + 56)/(2p) are perpendicular to each other, then find the value(s) of p.

Q18. Find the vector equation of a line passing through the origin and perpendicular to both the lines $r = 2i - j + 2k + \lambda(3i + 4j + 2k)$ and $r = \mu(i - j + k)$.

Q19. Vectors $a = 3i - 2j + 2k$ and $b = i + 2k$ represent the two adjacent sides of a parallelogram. Find the vectors representing its diagonals and hence find their lengths.

Q20. The area of the circle is increasing at a uniform rate of 2 cm^2/sec. How fast is the circumference of the circle increasing if the radius r = 5 cm?

Q21. Check whether the function f(x) defined as $f(x) = (x - 3)/[2(x - 3)]$, if $x < 3$, and $f(x) = (x - 6)/6$, if $x \geq 3$, is continuous at x = 3 or not.

Q22. Find the sub-interval of $(0, \infty)$ in which $f(x) = x^2e^{(-x)}$ is increasing.

Q23. Find: $\int (e^{(4x)} - 1)/(e^{(4x)} + 1)\, dx$

Q24. If M and m denote the local maximum and local minimum values of the function $f(x) = x + 1/x$, $x \neq 0$, respectively, find the value of (M - m).

Q25. Evaluate: $\tan^{-1}[2 \sin\{2 \cos^{-1}(\sqrt{3}/2)\}]$

Q26. A function $f : A \to B$ defined as $f(x) = 2x$ is both one-one and onto. If $A = \{1, 2, 3, 4\}$, then find the set B.

SECTION C: SHORT ANSWER QUESTIONS (SAQS)

Each question carries 3 marks. Provide clear analytical explanations or step-wise calculations.

Q27. If $I_1 = \int$ from $-\pi/4$ to $\pi/4$ [$dx/(1 + \cos 2x)$] and $I_2 = \int$ from $-1/2$ to $1/2$ $|x|\, dx$, then show that $I_1 - 4I_2 = 0$.

Q28. Evaluate: $\int$ from $-\pi/2$ to $\pi/2$ [$\cos^2 x/(2^x + 1)$] dx

Q29. The position vectors of vertices of ΔABC are A(2i - j + k), B(i - 3j - 5k) and C(3i - 4j - 4k). Find all the angles of ΔABC.

Q30. Verify that lines given by $r = (1 - \lambda)i + (\lambda - 2)j + (3 - 2\lambda)k$ and $r = (\mu + 1)i + (2\mu - 1)j - (2\mu + 1)k$ are skew lines. Hence, find shortest distance between the lines.

Q31. Solve the following linear programming problem graphically: Minimize $Z = 13x - 15y$, subject to constraints $x + y \leq 7$, $2x - 3y + 6 \geq 0$, $x \geq 0$, $y \geq 0$.

Q32. A survey was conducted on the patients who have undergone knee replacement surgeries. It was found that Robotic Knee replacement surgeries have 90% success rate. On a particular day, robotic surgery was performed on three patients, A, B and C, one after the other. Assuming that the success and failure of each surgery is independent of each other, find the probability that: (i) exactly one surgery is successful, (ii) at most two surgeries are successful.

Q33. The random variable X has the following probability distribution where a and b are some constants:
X: 1, 2, 3, 4, 5
P(X): 0.2, a, a, 0.2, b
If the mean E(X) = 3, then find values of a and b and hence determine $P(X \geq 3)$.

Q34. A pair of dice is thrown simultaneously. If X denotes the absolute difference of the numbers appearing on top of the dice, then find the probability distribution of X.

Q35. The side of an equilateral triangle is increasing at the rate of 3 cm/s. At what rate is area increasing when the side of the triangle is 15 cm?

Q36. Evaluate: $\int$ from 0 to $\pi/2$ [$\log(\sin x) - \log(2 \cos x)$] dx

SECTION D: LONG ANSWER QUESTIONS (LAQS)

Each question carries 5 marks. Comprehensive working, equations, diagrams and derivations may be required.

Q37. If $x = 3 \sin t - \sin 3t$, $y = 3 \cos t - \cos 3t$, find dy/dx and prove that $d^2y/dx^2 = -(\text{cosec}^3 2t \cdot \text{cosec } t)/3$.

Q38. Prove that the line through points A(0, -1, -1) and B(4, 5, 1) intersects the line through points C(3, 9, 4) and D(-4, 4, 4). Hence, write the equation of line passing through the point of intersection of lines AB and CD as well as origin.

Q39. Opposite sides of a square are along the lines: $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$. Find the area of the square if direction ratios of other pair of opposite sides of the square are given by $\langle -3, 6, p \rangle$. Also, find the value of p.

Q40. Find the image A' of the point A(1, 6, 3) in the line $x/1 = (y - 1)/2 = (z - 2)/3$. Also, find the equation of the line joining A and A'.

Q41. Solve the following system of equations, using matrices:
$2/x + 3/y + 10/z = 4$
$4/x - 6/y + 5/z = 1$
$6/x + 9/y - 20/z = 2$
where $x, y, z \neq 0$.

Q42. Find the absolute maximum and absolute minimum value of the function $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.

Q43. Solve the following system of equations by matrix method:
$x + 2y + 3z = 6$
$2x - y + z = 2$
$3x + 2y - 2z = 3$

Q44. If A_1 denotes the area of region bounded by $y^2 = 4x$, $x = 1$ and x-axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x$, $x = 4$, find $A_1 : A_2$.

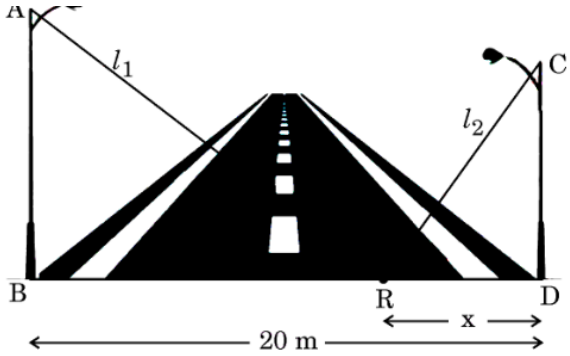
Q45. Solve the following L.P.P. graphically: Maximise $Z = 60x + 40y$ subject to $x + 2y \leq 12$, $2x + y \leq 12$, $4x + 5y \geq 20$, $x \geq 0$, $y \geq 0$.

Q46. Find the area of the region bounded by the curves $x^2 = y$, $y = x + 2$ and x-axis, using integration.

SECTION E: CASE STUDY-BASED QUESTIONS

Each question block carries 4 marks. Read the context carefully and answer all sub-questions that follow.

Q47. Two vertical light poles of height 22 m and 16 m stand on the opposite sides of a 20 m wide road as shown below in the figure. Two ladders of length l_1 and l_2 are placed from a common point R on the road at a distance of x m from the smaller pole.



Based on the above information, answer the following questions:
(i) Express $p(x) = l_1 + l_2$ in terms of x.
(ii) Find $p'(x)$.
(iii) (a) Find the value of x for which $l_1^2 + l_2^2$ is minimum.
OR
(iii) (b) If the 22 m long pole is also replaced by a 16 m long pole, at what distance from either pole should the ladders be kept so that the sum of squares of lengths of ladders needed to reach the top of the pole is minimum?

Q48. An online delivery company in a city has 5000 subscribers and collects annual subscription fees of ₹300 per subscriber for unlimited free deliveries. The company wishes to increase the annual subscription fee. It is predicted that, for every increase of ₹1, ten subscribers will discontinue. Assume that the company increased the annual fee by ₹x. Based on the given information, answer the following questions:
(i) How many subscribers will discontinue after an increase of ₹x in annual fee?
(ii) If R(x) denotes the total revenue collected after the increase of ₹x in subscription fee, express R(x) as a function of x.
(iii) (a) Find the value of x for which R(x) is maximum.
OR
(iii) (b) Find the sub-intervals of (0, 5000) in which R(x) is increasing and decreasing.

Q49. A survey was conducted to find out the success rate of students who qualified the entrance examination by dropping a year after class XII. As per the data collected, 40% students appearing in the examination were dropouts and the remaining students were regular students of class XII. Of the dropouts, 5% qualify the examination while 10% of the regular students qualify the examination. Based on the above information, answer the following questions:
(i) Find the probability that a student selected at random is a regular student.
(ii) A student is selected at random from a group of dropout students. What is the probability that the student will not qualify the examination?
(iii) (a) A student selected at random qualified the examination. Find the probability that student is not a dropout.
OR
(iii) (b) A student selected at random did not qualify the examination. Find the probability that the student was a regular student.

Q50. Overspeeding increases fuel consumption and decreases fuel economy as a result of tyre rolling friction and air resistance. While vehicles reach optimal fuel economy at different speeds, fuel mileage usually decreases rapidly at speeds above 80 km/h. The relation between fuel consumption F (l/100 km) and speed V (km/h) under some constraints is given as $F = V^2/500 - V/4 + 14$. On the basis of the above information, answer the following questions:
(i) Find F, when V = 40 km/h.
(ii) Find dF/dV.
(iii) (a) Find the speed V for which fuel consumption F is minimum.
OR
(iii) (b) Find the quantity of fuel required to travel 600 km at the speed V at which dF/dV = -0.01.

Q51. The month of September is celebrated as the Rashtriya Poshan Maah across the country. Following a healthy and well-balanced diet is crucial in order to supply the body with the proper nutrients it needs. A balanced diet also keeps us mentally fit and promotes improved level of energy. A dietician wishes to minimize the cost of a diet involving two types of foods, food X (x kg) and food Y (y kg), which are available at the rate of ₹16/kg and ₹20/kg respectively. The feasible region satisfying the constraints is shown in Figure-2. On the basis of the above information, answer the following questions:
(i) Identify and write all the constraints which determine the given feasible region in Figure-2.
(ii) If the objective is to minimize cost Z = 16x + 20y, find the values of x and y at which cost is minimum. Also, find minimum cost assuming that minimum cost is possible for the given unbounded region.

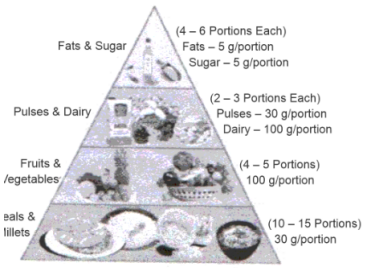


Figure-1

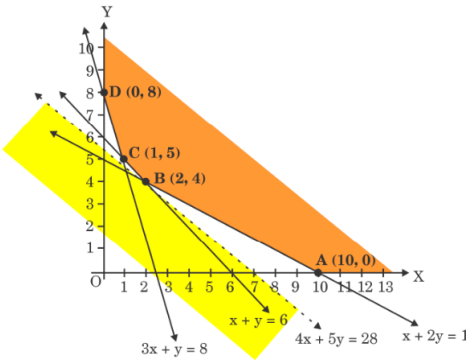
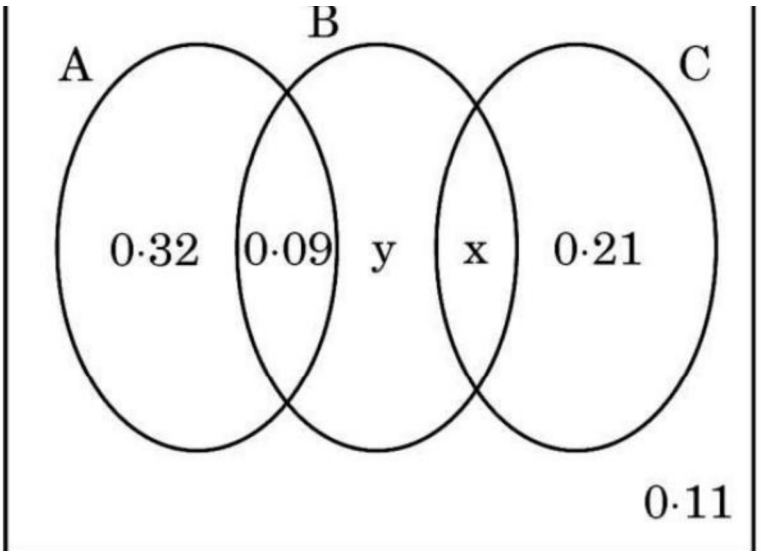


Figure-2

Q52. There are different types of Yoga which involve the usage of different poses of Yoga Asana, Meditation and Pranayama. In the Venn diagram, a society categorizes members using sets A, B and C. It is also given that the probability that a member does Yoga of type C is 0.44.



Based on the above information, answer the following questions:
(i) Find x.
(ii) Find y.
(iii) (a) Find P(C/B).
OR
(iii) (b) Verify that the probability that a randomly selected member of the society does Yoga of type A or B but not of type C is 0.41.